Scale Drawings

Lesson Plan

1. Lesson Information

Course	Grade 7 Accelerated (1205050)	Unit	5 — Multi-Step Proportional Relationships
Lesson	Lesson 1 — Scale Drawings	Date	_____
Length	53 minutes (one period)	Benchmark(s)	MA.7.GR.1.5

2. Standards

MA.7.GR.1.5 Solve mathematical and real-world problems involving dimensions and areas of geometric figures, including scale drawings and scale factors.

Must. Find the scale factor between two figures (k = copy ÷ original, in either direction of size) and use a scale factor or a scale to find a length in either direction; recognise that angles are unchanged. (MA.7.GR.1.5, clarifications 1 and 3.)

Must not. Perimeter and area under scaling are 5.02–03 (k^2); floor plans are 5.04; dilations on the coordinate plane are MA.8.GR.2.2; similar triangles by name are Unit 6.

3. Mathematical Thinking and Reasoning Standards

the evidence for each, from this period

MTR	What it asks	The evidence in this lesson
MA.K12.MTR.2.1	Demonstrate understanding by representing problems in multiple ways.	Notes I: the same pair of figures as a picture, two side lengths and one number k
MA.K12.MTR.4.1	Engage in discussions that reflect on the mathematical thinking of self and others.	boards 4 and 6 split; the sixty-second re-vote
MA.K12.MTR.5.1	Use patterns and structure to help understand and connect mathematical concepts.	boards 1–3: one question three times — k over 1, under 1, a fraction

4. Learning Target and Essential Question

Learning target. I can find the scale factor between two figures, and use a scale to find a real length from a drawing or a drawing length from the real one.

Essential question. If a copy's sides are 6 and 9 where the original's were 4 and 6, what one number was every side multiplied by?

Building on. Grade 6 (MA.6.AR.3.2) — a ratio as a unit rate; Unit 4 — multiplying and dividing decimals and fractions.

Working toward. 5.02–03 — perimeter by k, area by k^2; 5.04 — floor plans and maps; Unit 6 — similar triangles are scale copies.

5. Sequence

read from the deck; never typed

Segment	Min	What the teacher does	What students do
Settle in; post the learning target	1	Post the target; say the one line the Teacher's Edition gives for it, and nothing else yet.	Copy the target.
Warm-Up	6	Two minutes silent, then reveal. One sentence of reteach per question at most. Then the new words, read once.	Four retrieval questions, alone, no notes. Then copy the new words.
Notes	10	Teach from the front. Expand once per law; say the OFF line out loud.	Copy the notes; answer the checks aloud.
Example 1	3	Work the example; name the wrong boards before they appear.	Watch, then work the Your Turn alone.
Your Turn	3	Circulate; reveal and name what a wrong board most likely did.	Work it alone; boards up on the cue.
Example 2	3	Work the example; name the wrong boards before they appear.	Watch, then work the Your Turn alone.
Your Turn	3	Circulate; reveal and name what a wrong board most likely did.	Work it alone; boards up on the cue.
Whiteboards	13	One question at a time. Boards down until the cue, all up together; scan the back row first.	Answer on a board; hold it up on three.
Independent Set	6	Silent. Circulate and mark what you see; do not teach.	Six questions, written, alone.
IXL	5	Assign the day's IXL skills.	Start the skills; finish tonight.
Total	**53**		

6. Gradual Release

I do. The Notes slides and the worked half of each Example: the teacher writes, the class copies, and the OFF line is said out loud rather than printed.

We do. Each Example's Your Turn: the same steps on the student's own numbers, revealed and named a minute later.

You do. The nine-question whiteboard round (13 minutes) and then the six-question independent set, silent and written.

7. Higher-Order Questions

Question	DOK
Two rectangles are 4 by 6 and 8 by 10. Why is the second not a scale copy of the first?	DOK 3
A copy is smaller than its original. What do you know about k before you compute it?	DOK 2
Every side of a triangle is tripled. Why can none of its angles change?	DOK 3

8. Checks for Understanding

each board, and what to do about it

#	What the board asks	Answer	If the board is wrong
1	A 2 cm side becomes 6 cm in the copy. What is the scale factor?	***k = 3***	⅓ — divided original by copy [B1G-M 7.GR.1.5: the proportion set up backwards]; 4 — subtracted [the additive error]
2	An 8 cm side becomes 4 cm in the copy. What is the scale factor?	***k = ½***	2 — divided original by copy; 'k is always bigger than 1' [B1G-M 7.GR.1.5 named misconception]
3	A 12 cm side becomes 9 cm in the copy. What is the scale factor?	***k = ¾***	1⅓ — original ÷ copy [7.GR.1.5]; 3 — subtracted [the additive error]
4	A side of 5 cm becomes 2 cm in the copy. What is the scale factor?	***B — 0.4***	(A) original ÷ copy: 5 ÷ 2 [B1G-M 7.GR.1.5: the proportion set up backwards; 'k is always greater than 1']; (C) 'the same shape, so k = 1' [Math Nation 5.3: a scale factor of 1 is the same size, not the same shape]; (D) 5 − 2: the additive error — 'to get from 5 to 2, take away 3' [B1G-M 7.AR.3.3: 'just add' as a conversion]
5	The scale factor is 3.5. A side of the original is 6 cm. How long is that side in the copy?	***21 cm***	1.7 — divided by k going to the copy [Notes III: to the copy, multiply]
6	A drawing has a 40° angle. The real figure is a scale copy at k = 2.5. What is the angle in the real figure?	***A — 40°***	(B) 40 + 2.5: the additive error [B1G-M]; (C) 40 ÷ 2.5: the angle divided [7.GR.1.5]; (D) 40 × 2.5: the angle scaled like a length [7.GR.1.5: a scale factor multiplies lengths; the shape — its angles — is what stays the same]
7	A triangle has sides 6, 8 and 10. Another has sides 9, 12 and 15; a third has sides 9, 12 and 14. Which one is a scale copy of the first?	***The 9-12-15 triangle (k = 1.5); the 9-12-14 one is not a copy***	'both' — checked one pair and stopped [Notes I: the same k from every side]; 'neither' — compared 9 to 10 instead of corresponding sides
8	Dev, who is 13, draws a 12-ft wall as 3 in long. How long is a 20-ft wall on his drawing?	***5 in***	80 — multiplied by 4 going to the drawing [Notes III: back to the drawing, divide]; 11 — 20 − 12 + 3, the additive error
9	Priya says the scale factor from a 6 cm side to its 3 cm copy is 2. Is she right? Explain, and give the scale factor.	***No. k = 3 ÷ 6 = ½: the copy is smaller, so k is under 1.***	'Yes, because 6 is 2 times 3' — the proportion backwards [B1G-M 7.GR.1.5]; 'No, it is 3' — subtracted

MTR.4.1 — the one peer move. When a board splits — no option holding about two-thirds of the room — sixty seconds of "convince the person next to you", then re-vote, then reveal. The minute comes out of the whiteboard remainder.

9. Differentiation

ESE / 504. The direction is written on the board all period as one line — copy ÷ original — and every k is found by writing the copy's number over the original's before dividing. The two rectangles of Notes I stay up beside it.

ELL. **Scale factor** is one number with no unit; **scale** has units on it (1 in : 20 mi). Say "copy divided by original" with every k, and point to which figure is which.

Enrichment. Board 7 finishes early for some: ask them for a fourth triangle with sides that are a scale copy of 6-8-10 at k = 2.5, and whether its angles changed.

10. Closure, Vocabulary, Materials, Practice

Closure. Board 9 is written work and it is the exit evidence: the student has to say which way the division goes (copy ÷ original) and give k = ½. A board that says 'yes, 6 is 2 times 3' has the direction backwards — the unit's first misconception, and tomorrow's warm-up Q1 asks it again.

Vocabulary. **scale drawing**; **scale factor**; **corresponding sides**; **scale**

Materials. Whiteboards and markers. Two squares of different sizes cut from paper, for board 6's split: four right angles each. Calculators are allowed on everything and printed on nothing.

Practice. The six-question independent set is worked silently in class; whatever is not finished goes home. IXL, all skills required to a SmartScore of 67, due at the start of the next class: Scale drawings: find the scale factor, missing side, or missing angle — WEA.

Scale Factor: Perimeter and Area

Lesson Plan

1. Lesson Information

Course	Grade 7 Accelerated (1205050)	Unit	5 — Multi-Step Proportional Relationships
Lesson	Lessons 2–3 — Scale Factor: Perimeter and Area	Date	_____
Length	53 minutes (one period)	Benchmark(s)	MA.7.GR.1.5

2. Standards

MA.7.GR.1.5 Solve mathematical and real-world problems involving dimensions and areas of geometric figures, including scale drawings and scale factors.

Must. Find the perimeter and area of a scale copy from the original and k (perimeter × k, area × k^2), and find k from two areas when the ratio is a perfect square. (MA.7.GR.1.5, clarification 2.)

Must not. Area ratios that are not perfect squares (an approximate root); the area of a figure the class has no formula for; area and perimeter in different units (5.04 and 5.08–09).

3. Mathematical Thinking and Reasoning Standards

the evidence for each, from this period

MTR	What it asks	The evidence in this lesson
MA.K12.MTR.2.1	Demonstrate understanding by representing problems in multiple ways.	Notes II: the same copy as a grid picture, as 15 → 60, and as $k^2 = 4$
MA.K12.MTR.4.1	Engage in discussions that reflect on the mathematical thinking of self and others.	boards 4 and 6 split; the sixty-second re-vote on k against k^2
MA.K12.MTR.5.1	Use patterns and structure to help understand and connect mathematical concepts.	boards 1–2: the same k on the perimeter, then the area, and the room is asked why the second answer is bigger

4. Learning Target and Essential Question

Learning target. I can find the perimeter and the area of a scale copy from the original and k, and find k from two areas.

Essential question. Every side of a rectangle is doubled. Why is its area four times as big, not twice?

Building on. 5.01 — k = copy ÷ original; 4.05 — square roots of perfect squares; Grade 6 — perimeter and area of a rectangle.

Working toward. 5.04 — floor plans, where area has to be found in real units; 5.08–09 — area converts by the square of the length ratio.

5. Sequence

read from the deck; never typed

Segment	Min	What the teacher does	What students do
Settle in; post the learning target	1	Post the target; say the one line the Teacher's Edition gives for it, and nothing else yet.	Copy the target.
Warm-Up	6	Two minutes silent, then reveal. One sentence of reteach per question at most. Then the new words, read once.	Four retrieval questions, alone, no notes. Then copy the new words.
Notes	10	Teach from the front. Expand once per law; say the OFF line out loud.	Copy the notes; answer the checks aloud.
Example 1	4	Work the example; name the wrong boards before they appear.	Watch, then work the Your Turn alone.
Your Turn	3	Circulate; reveal and name what a wrong board most likely did.	Work it alone; boards up on the cue.
Example 2	4	Work the example; name the wrong boards before they appear.	Watch, then work the Your Turn alone.
Your Turn	3	Circulate; reveal and name what a wrong board most likely did.	Work it alone; boards up on the cue.
Whiteboards	11	One question at a time. Boards down until the cue, all up together; scan the back row first.	Answer on a board; hold it up on three.
Independent Set	6	Silent. Circulate and mark what you see; do not teach.	Six questions, written, alone.
IXL	5	Assign the day's IXL skills.	Start the skills; finish tonight.
Total	**53**		

6. Gradual Release

I do. The Notes slides and the worked half of each Example: the teacher writes, the class copies, and the OFF line is said out loud rather than printed.

We do. Each Example's Your Turn: the same steps on the student's own numbers, revealed and named a minute later.

You do. The nine-question whiteboard round (11 minutes) and then the six-question independent set, silent and written.

7. Higher-Order Questions

Question	DOK
A rectangle's sides are tripled. Its perimeter is multiplied by 3 and its area by 9. Why are the two numbers different?	DOK 3
Two squares have areas 25 and 100. What is k, and how do you know without finding a side?	DOK 2
Could a copy have the same perimeter as its original and a different area? Explain.	DOK 3

8. Checks for Understanding

each board, and what to do about it

#	What the board asks	Answer	If the board is wrong
1	A figure with perimeter 14 cm is scaled by k = 2. What is the copy's perimeter?	***28 cm***	56 — perimeter × k^2 [Notes I: a perimeter is lengths, × k]; 16 — added k [the additive error]
2	A figure with area 12 cm^2 is scaled by k = 2. What is the copy's area?	***48 cm^2***	24 — area × k [B1G-M 7.GR.1.5 named misconception: scaling area like length]
3	A figure with area 48 cm^2 is scaled by k = ½. What is the copy's area?	***12 cm^2***	24 — area × k [7.GR.1.5: area × k^2]; 96 — divided by ½ as if k were 2 [5.01: k under 1 shrinks]
4	A figure with area 10 cm^2 is scaled by k = 3. What is the copy's area?	***A — 90 cm^2***	(B) area × 2k: 'two lengths, so twice k' [Notes II: k × k, not k + k]; (C) area × k: the guide's named misconception [B1G-M 7.GR.1.5: scaling area like length]; (D) 10 + 3: the additive error [B1G-M]
5	A figure with area 8 cm^2 is scaled by k = 1.5. What is the copy's area?	***18 cm^2***	12 — area × k [7.GR.1.5]; 24 — area × 2k [Notes II: k × k]
6	A figure's perimeter is 20 cm; its copy's perimeter is 60 cm. What is the area ratio, copy to original?	***D — 9***	(A) 2k: 'area has two dimensions, so double it' [Notes II]; (B) the perimeter ratio handed in as the area ratio [Notes I and II: perimeter gives k, area is k^2]; (C) 60 − 20: the additive error [B1G-M]
7	A triangle with sides 6, 8 and 10 cm is scaled by k = ½. Its area is 24 cm^2. Find the copy's perimeter and area.	***P = 12 cm, A = 6 cm^2***	P 12, A 12 — the area halved like a length [7.GR.1.5]; P 6, A 6 — both quartered
8	A 4 in by 6 in photo, taken 3 years ago, is enlarged at k = 2.5. What is the enlargement's area?	***150 in^2***	60 — area × k [7.GR.1.5]; 120 — 24 × 5, k doubled [Notes II]
9	Jo says that doubling every side of a rectangle doubles its area. Explain why not, with a counterexample.	***No: 2 by 7 doubled is 4 by 14, area 14 → 56 — × 4, as both lengths doubled.***	'Yes, because every side doubles' — scaling area like length [B1G-M 7.GR.1.5]; 'No, it is k^{2}' with no numbers — the rule without the reason

MTR.4.1 — the one peer move. When a board splits — no option holding about two-thirds of the room — sixty seconds of “convince the person next to you”, then re-vote, then reveal. The minute comes out of the whiteboard remainder.

9. Differentiation

ESE / 504. Two lines stay on the board all period: perimeter × k, area × k × k. Every board is answered by pointing at one of them before multiplying. The grid picture of Notes II stays up beside them.

ELL. **Perimeter** is the distance around; **area** is the squares inside. Say "around" and "inside" with each board, and the word **squared** means times itself.

Enrichment. Board 7 finishes early for some: ask them what k does to a triangle's angles (nothing) and to its height (× k), and why the area rule still holds when there is no length × width.

10. Closure, Vocabulary, Materials, Practice

Closure. Board 9 is written work and it is the exit evidence: a sentence that area is two lengths multiplied, so it is multiplied by k twice, with a counterexample's numbers (a 2 by 7 at k = 2: 14 → 56, not 28). A board that only says 'because k^{2}' has named the rule, not the reason.

Vocabulary. **k squared**; **perimeter ratio**; **area ratio**

Materials. Whiteboards and markers. A sheet of grid paper under the document camera for Notes II's four-copies picture, drawn live if the slide's grid is not enough. Calculators are allowed on everything and printed on nothing.

Practice. The six-question independent set is worked silently in class; whatever is not finished goes home. IXL, all skills required to a SmartScore of 67, due at the start of the next class: Perimeter and area: changes in scale — ZC6; Scale drawings: find the scale factor or missing side — XHM.

Real-World Scale Drawings

Lesson Plan

1. Lesson Information

Course	Grade 7 Accelerated (1205050)	Unit	5 — Multi-Step Proportional Relationships
Lesson	Lesson 4 — Real-World Scale Drawings	Date	_____
Length	53 minutes (one period)	Benchmark(s)	MA.7.GR.1.5

2. Standards

MA.7.GR.1.5 Solve mathematical and real-world problems involving dimensions and areas of geometric figures, including scale drawings and scale factors.

Must. Read a scale such as ¼ in : 1 ft or 1 : 50 as a rate, find real lengths and drawing lengths from it in both directions, and find a real area from the real dimensions. (MA.7.GR.1.5, clarifications 1 and 3; B1G-M's wall example.)

Must not. Converting the real area into another system (5.08–09); drawing a plan to scale with a ruler (not a FAST shape — the audit's ruler items are out); a scale factor between two drawings of one object (5.01).

3. Mathematical Thinking and Reasoning Standards

the evidence for each, from this period

MTR	What it asks	The evidence in this lesson
MA.K12.MTR.1.1	Actively participate in effortful learning both individually and collectively.	Example 2 and its Your Turn: a distance, then a time — the first problem that does not stop at the first answer
MA.K12.MTR.4.1	Engage in discussions that reflect on the mathematical thinking of self and others.	boards 4 and 6 split; the sixty-second re-vote
MA.K12.MTR.7.1	Apply mathematics to real-world contexts.	every board is a plan, a map or a model: the scale is a real object's, and the answer is checked against what a room or a car could be

4. Learning Target and Essential Question

Learning target. I can use a scale such as ¼ in : 1 ft to find real lengths and areas from a drawing, and drawing lengths from real ones.

Essential question. A room is 2 in by 3 in on a plan at 1 in : 4 ft. Why is its real area not 6 × 4 square feet?

Building on. 5.01 — a map scale used both ways; 5.02–03 — area is two lengths, so the rate is used twice; Grade 6 — a ratio written as a unit rate.

Working toward. 5.06–07 — a conversion ratio is a rate too, and it is multiplied the same way; 5.10–11 — a scale, then a speed: two rates in one problem.

5. Sequence

read from the deck; never typed

Segment	Min	What the teacher does	What students do
Settle in; post the learning target	1	Post the target; say the one line the Teacher's Edition gives for it, and nothing else yet.	Copy the target.
Warm-Up	6	Two minutes silent, then reveal. One sentence of reteach per question at most. Then the new words, read once.	Four retrieval questions, alone, no notes. Then copy the new words.
Notes	10	Teach from the front. Expand once per law; say the OFF line out loud.	Copy the notes; answer the checks aloud.
Example 1	4	Work the example; name the wrong boards before they appear.	Watch, then work the Your Turn alone.
Your Turn	3	Circulate; reveal and name what a wrong board most likely did.	Work it alone; boards up on the cue.
Example 2	4	Work the example; name the wrong boards before they appear.	Watch, then work the Your Turn alone.
Your Turn	3	Circulate; reveal and name what a wrong board most likely did.	Work it alone; boards up on the cue.
Whiteboards	11	One question at a time. Boards down until the cue, all up together; scan the back row first.	Answer on a board; hold it up on three.
Independent Set	6	Silent. Circulate and mark what you see; do not teach.	Six questions, written, alone.
IXL	5	Assign the day's IXL skills.	Start the skills; finish tonight.
Total	**53**		

6. Gradual Release

I do. The Notes slides and the worked half of each Example: the teacher writes, the class copies, and the OFF line is said out loud rather than printed.

We do. Each Example's Your Turn: the same steps on the student's own numbers, revealed and named a minute later.

You do. The nine-question whiteboard round (11 minutes) and then the six-question independent set, silent and written.

7. Higher-Order Questions

Question	DOK
A plan's scale is ¼ in : 1 ft. Why does writing it as 1 in : 4 ft make every question easier?	DOK 2
A room is 2 in by 3 in on that plan. Why is its real area 96 ft^2 and not 24?	DOK 3
Two models of one plane are built at 1 : 48 and 1 : 72. Which is longer, and why does the bigger number make the smaller model?	DOK 3

8. Checks for Understanding

each board, and what to do about it

#	What the board asks	Answer	If the board is wrong
1	A plan is drawn at ½ in : 1 ft. A wall is 4 in long on it. How long is the real wall?	***8 ft***	2 — divided (4 × ½) [Notes I: per 1 inch first, then × 2]; 4 — read the scale as 1 : 1
2	A plan is drawn at 1 in : 3 ft. A room is 2.5 in long on it. How long is the real room?	***7.5 ft***	5.5 — added 3 [B1G-M 7.GR.1.5: the additive error]; 0.83 — divided [Notes I: drawing to real is × rate]
3	A plan is drawn at 1 in : 3 ft. A real wall is 21 ft long. How long is it on the plan?	***7 in***	63 — multiplied [Notes III: real to drawing is a division]; 18 — subtracted 3
4	A model car is built at 1 : 50 and is 8 cm long. How long is the real car?	***A — 4 m***	(B) 8 ÷ 50: divided [Notes I: drawing to real is × rate]; (C) 400 with the wrong unit: 400 cm, not 400 m [Notes I: the scale's units are the drawing's]; (D) 8 + 50: the additive error [B1G-M 7.GR.1.5]
5	A plan is drawn at ¼ in : 1 ft. A real wall is 15 ft long. How long is it on the plan?	***3.75 in***	60 — multiplied by 4 [Notes III: real to drawing is ÷]; 7.5 — divided by 2
6	A plan is drawn at 1 in : 4 ft. A room is 2 in by 3 in on it. What is the room's real area?	***B — 96 ft²***	(A) 6 in² × 4: the drawing's area scaled by the length rate [B1G-M 7.GR.1.5: scaling area like length; Notes II]; (C) 2(8 + 12): the real perimeter, not the area [5.02–03 Notes I]; (D) 8 + 12: the real dimensions added [B1G-M 7.GR.1.5: the additive error]
7	A plane is 48 ft long. One model of it is built at 1 : 48, another at 1 : 72. Which model is longer, and by how much?	***The 1 : 48 model, by 4 in (12 in against 8 in)***	the 1 : 72 model — 'the bigger scale is the bigger model' [Notes III: real ÷ rate — a bigger rate divides more]; 1 ft and 0.67 ft with no comparison — stopped early
8	Mia, who is 12, enlarges her 9 in by 12 in poster into a mural at 1 in : 2 ft. What are the mural's dimensions?	***18 ft by 24 ft***	4.5 ft by 6 ft — divided [Notes I: drawing to real is × rate]; 18 in by 24 in — kept the drawing's unit
9	Andrew's drawing shows a wall 3.5 in wide; the real wall is 26.25 ft wide. The drawn height is 4 in. Find the real height and the wall's real area, and say which step uses the scale factor.	***30 ft high; 787.5 ft². The scale factor (7.5 ft per in) is the step from 4 in to 30 ft.***	105 ft² — 3.5 × 4 = 14 in², then × 7.5 [B1G-M 7.GR.1.5: scaling area like length]; 'the area step is the scale factor' — the scale is between a drawing and a real length, not two real ones

MTR.4.1 — the one peer move. When a board splits — no option holding about two-thirds of the room — sixty seconds of “convince the person next to you”, then re-vote, then reveal. The minute comes out of the whiteboard remainder.

9. Differentiation

ESE / 504. Every scale is rewritten per 1 drawing unit on the board before a question is read, and stays there. Two lines under it all period: drawing × rate = real; real ÷ rate = drawing.

ELL. **Scale** here is a ratio, not the kitchen scale; **plan** means a drawing from above. Say "on the paper" for drawing lengths and "in the room" for real ones with every board.

Enrichment. Board 7 finishes early for some: ask them to write the two models' lengths in centimetres with 1 in = 2.54 cm — Monday's lesson, a day early.

10. Closure, Vocabulary, Materials, Practice

Closure. Board 9 is written work and it is the exit evidence: the scale read as a rate (7.5 ft per inch), the height from it (30 ft), the area from the two real dimensions (787.5 ft^2), and a sentence naming which step is the scale factor. A board that scales $3.5 \times 4 = 14$ in^2 by 7.5 has not gone to real units first.

Vocabulary. **floor plan**; **scale (two units)**; **unit rate of a scale**

Materials. Whiteboards and markers. A printed floor plan under the document camera for Notes II if the slide's rectangle is not enough. Calculators are allowed on everything and printed on nothing.

Practice. The six-question independent set is worked silently in class; whatever is not finished goes home. IXL, all skills required to a SmartScore of 67, due at the start of the next class: Scale drawings: word problems — 84H; Scale drawings: scale factor word problems — KCM.

Converting Length Across Systems; Currency

Lesson Plan

1. Lesson Information

Course	Grade 7 Accelerated (1205050)	Unit	5 — Multi-Step Proportional Relationships
Lesson	Lessons 6–7 — Converting Length Across Systems; Currency	Date	_____
Length	53 minutes (one period)	Benchmark(s)	MA.7.AR.3.3

2. Standards

MA.7.AR.3.3 Solve mathematical and real-world problems involving the conversion of units across different measurement systems.

Must. Convert a length between customary and metric with a printed reference-sheet ratio, either direction, in a chain; convert money with a given rate, both ways. (MA.7.AR.3.3.)

Must not. A ratio not on the FAST reference sheet; a real exchange rate the student must know; mass, volume or area (5.08–09).

3. Mathematical Thinking and Reasoning Standards

the evidence for each, from this period

MTR	What it asks	The evidence in this lesson
MA.K12.MTR.6.1	Assess the reasonableness of solutions.	Notes II and board 6: the estimate before the arithmetic — a centimetre is shorter than an inch, so fewer inches; a yard is nearly a metre, so about 270 cm
MA.K12.MTR.4.1	Engage in discussions that reflect on the mathematical thinking of self and others.	boards 4 and 6 split; the sixty-second re-vote on which ratio
MA.K12.MTR.3.1	Complete tasks with mathematical fluency.	boards 5–6: two conversions in a row, each one a form of 1, with the units cancelling on the paper

4. Learning Target and Essential Question

Learning target. I can convert a length between the customary and metric systems with a ratio from the reference sheet, and convert money with a given rate.

Essential question. 1 in = 2.54 cm and 1 cm = 0.39 in are both on the sheet. How do you know which one to multiply by?

Building on. 5.04 — a scale read as a rate; 5.01 — multiplying by a rate; Grade 6 — converting inside one system (12 in = 1 ft).

Working toward. 5.08–09 — the same move on mass and volume, and squared for area; 5.10–11 — a conversion inside a rate comparison.

5. Sequence

read from the deck; never typed

Segment	Min	What the teacher does	What students do
Settle in; post the learning target	1	Post the target; say the one line the Teacher's Edition gives for it, and nothing else yet.	Copy the target.
Warm-Up	6	Two minutes silent, then reveal. One sentence of reteach per question at most. Then the new words, read once.	Four retrieval questions, alone, no notes. Then copy the new words.
Notes	10	Teach from the front. Expand once per law; say the OFF line out loud.	Copy the notes; answer the checks aloud.
Example 1	4	Work the example; name the wrong boards before they appear.	Watch, then work the Your Turn alone.
Your Turn	3	Circulate; reveal and name what a wrong board most likely did.	Work it alone; boards up on the cue.
Example 2	4	Work the example; name the wrong boards before they appear.	Watch, then work the Your Turn alone.
Your Turn	3	Circulate; reveal and name what a wrong board most likely did.	Work it alone; boards up on the cue.
Whiteboards	11	One question at a time. Boards down until the cue, all up together; scan the back row first.	Answer on a board; hold it up on three.
Independent Set	6	Silent. Circulate and mark what you see; do not teach.	Six questions, written, alone.
IXL	5	Assign the day's IXL skills.	Start the skills; finish tonight.
Total	**53**		

6. Gradual Release

I do. The Notes slides and the worked half of each Example: the teacher writes, the class copies, and the OFF line is said out loud rather than printed.

We do. Each Example's Your Turn: the same steps on the student's own numbers, revealed and named a minute later.

You do. The nine-question whiteboard round (11 minutes) and then the six-question independent set, silent and written.

7. Higher-Order Questions

Question	DOK
Why does multiplying 15 in by 2.54 cm per inch change the unit and not the length?	DOK 2
Jo turned 30 cm into 76.2 in. Without computing, how do you know that is wrong?	DOK 3
Money converts like length, but the ratio is on the problem and not on the sheet. Why?	DOK 2

8. Checks for Understanding

each board, and what to do about it

#	What the board asks	Answer	If the board is wrong
1	1 in = 2.54 cm. A ruler is 20 in long. How many centimetres is that?	***50.8 cm***	7.87 — divided by 2.54 [Notes II: centimetres are smaller, so more of them]; 22.54 — added the ratio
2	1 cm = 0.39 in. A phone is 20 cm long. How many inches is that?	***7.8 in***	50.8 — multiplied by 2.54 [Notes II: pick the ratio whose 1 is the unit you have]; 0.78 — place value
3	1 mi = 1.61 km. A race is 5 mi long. How many kilometres is that?	***8.05 km***	3.1 — multiplied by 0.62, the other ratio [Notes II]; 6.61 — added 1.61
4	1 km = 0.62 mi. A trail is 5 km long. How many miles is that?	***D — 3.1 mi***	(A) 5 + 0.62: the additive error [B1G-M]; (B) place value: 5 × 0.62 is 3.1, not 0.31 [4.03]; (C) × 1.61, the ratio for the other direction [B1G-M 7.AR.3.3: the inverted ratio; Notes II]
5	1 ft = 12 in and 1 in = 2.54 cm. A door is 6 ft tall. How many centimetres is that?	***182.88 cm***	15.24 — 6 × 2.54, the feet treated as inches [Notes I: the unit you have must match the ratio's 1]; 183 — rounded early (accept with the work shown)
6	1 yd = 3 ft, 1 ft = 12 in, 1 in = 2.54 cm. A table is 3 yd long. About how many centimetres is that? Estimate, then compute.	***About 270; exactly 274.32 cm***	7.62 — 3 × 2.54, the yards treated as inches [Notes II: estimate first — a 3-yd table is not 8 cm]; 91.44 — stopped at 36 in × 2.54 (one yard)
7	Rates on 2 Nov: $1 = 1.32 CAD; 1 CAD = 16.2 pesos. Theo changes $200 to CAD, then to pesos. How many pesos does he get?	***4,276.80 pesos***	3,240 — 200 × 16.2, the first hop skipped [Notes III: the rate's 1 must be the currency you have]; 2,454.55 — divided by 1.32 on the first hop
8	1 km = 0.62 mi. Theo, who weighs 110 lb, runs a 5 km race. How long is the race in miles?	***3.1 mi***	8.05 — × 1.61 [B1G-M 7.AR.3.3: the inverted ratio]; 49.94 kg — converted the weight instead
9	1 in = 2.54 cm and 1 cm = 0.39 in. Jo converts 30 cm to inches by multiplying by 2.54 and gets 76.2 in. Name the error and fix it.	***She used the inches ratio on centimetres. Inches are bigger, so fewer: 11.7 in.***	'She should have used 0.39' with no reason — the fix without the error named; 76.2 defended as 'more inches' — the estimate backwards [B1G-M 7.AR.3.3]

MTR.4.1 — the one peer move. When a board splits — no option holding about two-thirds of the room — sixty seconds of “convince the person next to you”, then re-vote, then reveal. The minute comes out of the whiteboard remainder.

9. Differentiation

ESE / 504. The reference sheet's two conversion lines stay on the board all period. Every board starts by pointing at the ratio whose 1 is the unit you have, then multiplying. The estimate is said aloud before the calculator.

ELL. **Customary** is the system with inches and miles; **metric** has centimetres and kilometres. The word **per** in "2.54 cm per inch" is the same per as miles per hour.

Enrichment. Board 7 finishes early for some: ask them to run the chain backwards — pesos to Canadian dollars to US dollars — and say which operation each step became.

10. Closure, Vocabulary, Materials, Practice

Closure. Board 9 is written work and it is the exit evidence: the error named (Jo multiplied by the ratio whose 1 is inches, when she had centimetres), the estimate that catches it (inches are longer, so fewer of them), and the fix (30 × 0.39 = 11.7 in). A board that writes the right number with no sentence about the direction has not met the lesson.

Vocabulary. **customary**; **metric**; **conversion ratio**; **exchange rate**

Materials. Whiteboards and markers. The FAST reference sheet's conversion lines on the board all period (they are printed on every item as well). Calculators are allowed on everything and printed on nothing.

Practice. The six-question independent set is worked silently in class; whatever is not finished goes home. IXL, all skills required to a SmartScore of 67, due at the start of the next class: Convert between customary and metric systems — E8Z; Compare and convert customary units — 4B8; Compare and convert metric units — HW2.

Converting Mass, Area and Volume

Lesson Plan

1. Lesson Information

Course	Grade 7 Accelerated (1205050)	Unit	5 — Multi-Step Proportional Relationships
Lesson	Lessons 8–9 — Converting Mass, Area and Volume	Date	_____
Length	53 minutes (one period)	Benchmark(s)	MA.7.AR.3.3

2. Standards

MA.7.AR.3.3 Solve mathematical and real-world problems involving the conversion of units across different measurement systems.

Must. Convert mass and volume with a printed reference-sheet ratio, either direction; convert area by squaring the length ratio (and volume by cubing it, in writing); route fluid ounces through cups. (MA.7.AR.3.3.)

Must not. A ratio not on the FAST reference sheet (no fl oz → mL, no ft^2 → m^2 as one number); a density or a weight-from-volume question; the capacity of a solid from its dimensions (Unit 6).

3. Mathematical Thinking and Reasoning Standards

the evidence for each, from this period

MTR	What it asks	The evidence in this lesson
MA.K12.MTR.5.1	Use patterns and structure to help understand and connect mathematical concepts.	Notes II: yesterday's k^2 and today's 144 are the same structure, and the room is asked to say so
MA.K12.MTR.4.1	Engage in discussions that reflect on the mathematical thinking of self and others.	boards 4 and 6 split; the sixty-second re-vote
MA.K12.MTR.6.1	Assess the reasonableness of solutions.	every mass and volume board says its estimate first — fewer kilograms than pounds, more litres than gallons

4. Learning Target and Essential Question

Learning target. I can convert mass, volume and area between systems, and square the length ratio for area.

Essential question. 1 ft = 12 in. Why is 1 ft^2 not 12 in^2?

Building on. 5.06–07 — multiply by the sheet's ratio, estimate first; 5.02–03 — area is two lengths, so k is used twice; Grade 6 — 1 ft^2 is a 12 by 12 square.

Working toward. 5.10–11 — a conversion inside a multi-step problem; Unit 6 — surface area and volume in whichever unit the problem names.

5. Sequence

read from the deck; never typed

Segment	Min	What the teacher does	What students do
Settle in; post the learning target	1	Post the target; say the one line the Teacher's Edition gives for it, and nothing else yet.	Copy the target.
Warm-Up	6	Two minutes silent, then reveal. One sentence of reteach per question at most. Then the new words, read once.	Four retrieval questions, alone, no notes. Then copy the new words.
Notes	10	Teach from the front. Expand once per law; say the OFF line out loud.	Copy the notes; answer the checks aloud.
Example 1	4	Work the example; name the wrong boards before they appear.	Watch, then work the Your Turn alone.
Your Turn	3	Circulate; reveal and name what a wrong board most likely did.	Work it alone; boards up on the cue.
Example 2	4	Work the example; name the wrong boards before they appear.	Watch, then work the Your Turn alone.
Your Turn	3	Circulate; reveal and name what a wrong board most likely did.	Work it alone; boards up on the cue.
Whiteboards	11	One question at a time. Boards down until the cue, all up together; scan the back row first.	Answer on a board; hold it up on three.
Independent Set	6	Silent. Circulate and mark what you see; do not teach.	Six questions, written, alone.
IXL	5	Assign the day's IXL skills.	Start the skills; finish tonight.
Total	**53**		

6. Gradual Release

I do. The Notes slides and the worked half of each Example: the teacher writes, the class copies, and the OFF line is said out loud rather than printed.

We do. Each Example's Your Turn: the same steps on the student's own numbers, revealed and named a minute later.

You do. The nine-question whiteboard round (11 minutes) and then the six-question independent set, silent and written.

7. Higher-Order Questions

Question	DOK
1 m = 100 cm. Why is 1 m^2 = 10,000 cm^2 and not 100?	DOK 3
A 5-lb bag and a 2-kg bag: which is heavier, and what has to happen before you can say?	DOK 2
Why do fluid ounces go through cups to reach litres on this reference sheet?	DOK 2

8. Checks for Understanding

each board, and what to do about it

#	What the board asks	Answer	If the board is wrong
1	1 kg = 2.204 lb. A bag of rice is 10 kg. How many pounds is that?	***22.04 lb***	4.54 — × 0.454, the other ratio [B1G-M 7.AR.3.3: the inverted ratio]; 12.204 — added
2	1 gal = 3.785 L. A cooler holds 4 gal. How many litres is that?	***15.14 L***	1.06 — × 0.264 [B1G-M 7.AR.3.3: the inverted ratio]; 7.785 — added
3	1 L = 0.264 gal. A tank holds 10 L. How many gallons is that?	***2.64 gal***	37.85 — × 3.785 [Notes I: pick the ratio whose 1 is the unit you have]; 0.264 — forgot the 10
4	1 yd = 3 ft. A patch of lawn is 2 yd^2 in area. How many square feet is that?	***B — 18 ft^2***	(A) 2 × 6: 'two dimensions, so double the ratio' [5.02–03 Notes II: k × k, not 2k]; (C) 3^2 alone: the 2 yd^2 forgotten — one square yard, not two [4.06: finish the computation]; (D) 2 × 3: the ratio used once [B1G-M 7.AR.3.3: square units need the ratio squared; Notes II]
5	1 c = 8 fl oz and 1 c = 0.24 L. A bottle holds 16 fl oz. How many litres is that?	***0.48 L***	3.84 — 16 × 0.24, the fluid ounces treated as cups [Notes III: the sheet's path is fl oz → c → L]; 453.6 — used 28.35, the weight ounce [vocabulary: oz and fl oz are two units]
6	1 m = 100 cm. How many square centimetres is 1 m^2?	***C — 10,000 cm^2***	(A) 10 × 100: the ratio once, then a guess upward [B1G-M 7.AR.3.3: square units need the ratio squared]; (B) 2 × 100: 'two dimensions, so double it' [5.02–03 Notes II]; (D) the length ratio handed in for an area [B1G-M 7.AR.3.3: square units need the ratio squared; Notes II]
7	1 kg = 2.204 lb. A 5-lb bag of flour and a 2-kg bag of sugar. Which is heavier, and by how much?	***The 5-lb bag, by 0.592 lb (about 0.6 lb)***	'the 5-lb bag, because 5 > 2' — the numbers compared, not the weights [MTR.6.1: units first]; 'the 2-kg bag' — kilograms taken as heavier than any number of pounds
8	1 gal = 3.785 L. A 20-gal fish tank, bought for $45, is being filled. How many litres does it hold?	***75.7 L***	5.28 — × 0.264 [B1G-M 7.AR.3.3: the inverted ratio]; 170.33 — the price used
9	1 ft = 12 in. Priya says 2 ft^3 is 24 in^3, because 2 × 12 = 24. Name the error and fix it.	***She used the ratio once. A cubic foot is 12 × 12 × 12 = 1,728 in^3, so 2 ft^3 = 3,456 in^3.***	288 in^3 — the ratio squared, today's rule applied to a cube [Notes III: three lengths, three times]; 'it should be 24 ft' — the unit changed instead of the number

MTR.4.1 — the one peer move. When a board splits — no option holding about two-thirds of the room — sixty seconds of “convince the person next to you”, then re-vote, then reveal. The minute comes out of the whiteboard remainder.

9. Differentiation

ESE / 504. Three lines on the board all period: length — the ratio once; area — twice; volume — three times. Every board is answered by pointing at one line first. A 12 by 12 grid stays up beside them.

ELL. **Mass** and **weight** are one word here; **ounce** and **fluid ounce** are two different things — say "weight ounce" and "liquid ounce" aloud. **Square** means two lengths; **cubic** means three.

Enrichment. Board 9 finishes early for some: ask for 1 m^3 in cm^3 (1,000,000) and whether 1 L = 1,000 cm^3 makes sense from it (1 L = 1,000 mL; a litre is a 10-cm cube).

10. Closure, Vocabulary, Materials, Practice

Closure. Board 9 is written work and it is the exit evidence: the error named (Priya used the ratio once, and a cubic foot is three lengths), and the fix ($12 \times 12 \times 12 = 1{,}728$ in^3 per ft^3, so 2 ft^3 = 3,456 in^3). A board that writes 3,456 with no sentence about three lengths has named the number, not the reason.

Vocabulary. **mass and weight**; **ounce (oz)**; **fluid ounce (fl oz)**; **square units**; **cubic units**

Materials. Whiteboards and markers. The FAST reference sheet's conversion lines on the board all period (printed on every item as well); a 12 by 12 grid under the document camera for Notes II. The Unit Review paper to send home. Calculators are allowed on everything and printed on nothing.

Practice. The six-question independent set is worked silently in class; whatever is not finished goes home. IXL, all skills required to a SmartScore of 67, due at the start of the next class: Convert between customary and metric systems — E8Z; Compare and convert customary units — 4B8; Compare and convert metric units — HW2.

Multi-Step Proportion Problems

Lesson Plan

1. Lesson Information

Course	Grade 7 Accelerated (1205050)	Unit	5 — Multi-Step Proportional Relationships
Lesson	Lessons 10–11 — Multi-Step Proportion Problems	Date	_____
Length	53 minutes (one period)	Benchmark(s)	MA.7.AR.4.5

2. Standards

MA.7.AR.4.5 Solve real-world problems involving proportional relationships.

Must. Solve a two- or three-step proportional problem (a rate used twice; a recipe scaled and checked; a part-to-whole mixture), and compare two rates after putting them in one unit with a printed reference-sheet ratio. (MA.7.AR.4.5, with MA.7.AR.3.3 inside it.)

Must not. A graph, a table of a proportional relationship, or y = px named as such (MA.7.AR.4.1–4.4, Unit 7); a ratio not on the FAST sheet; a percent problem (Unit 8).

3. Mathematical Thinking and Reasoning Standards

the evidence for each, from this period

MTR	What it asks	The evidence in this lesson
MA.K12.MTR.1.1	Actively participate in effortful learning both individually and collectively.	Notes I and board 4: a problem whose first answer (miles per tank) is not the answer, and the room keeps going
MA.K12.MTR.6.1	Assess the reasonableness of solutions.	Notes II and board 6: 'which is faster?' is refused until the units match; the answer is checked against the estimate that a mile is longer
MA.K12.MTR.7.1	Apply mathematics to real-world contexts.	boards 8–9: a plan's area in square metres and a choice of car — the unit's whole toolkit on one real question

4. Learning Target and Essential Question

Learning target. I can solve a problem that takes two or more proportional steps, including a unit conversion, and compare two rates given in different units.

Essential question. A car does 100 km/h and another does 65 mi/h. Why can't you say which is faster yet?

Building on. 5.06–09 — the sheet's ratios; 5.04 — a scale, then a speed: two rates in one problem; Grade 6 — a unit rate.

Working toward. Unit 7 — the constant of proportionality as a graph and an equation; the Unit 5 test's two transfer items, which are boards 8 and 9 in a third context.

5. Sequence

read from the deck; never typed

Segment	Min	What the teacher does	What students do
Settle in; post the learning target	1	Post the target; say the one line the Teacher's Edition gives for it, and nothing else yet.	Copy the target.
Warm-Up	6	Two minutes silent, then reveal. One sentence of reteach per question at most. Then the new words, read once.	Four retrieval questions, alone, no notes. Then copy the new words.
Notes	10	Teach from the front. Expand once per law; say the OFF line out loud.	Copy the notes; answer the checks aloud.
Example 1	4	Work the example; name the wrong boards before they appear.	Watch, then work the Your Turn alone.
Your Turn	3	Circulate; reveal and name what a wrong board most likely did.	Work it alone; boards up on the cue.
Example 2	4	Work the example; name the wrong boards before they appear.	Watch, then work the Your Turn alone.
Your Turn	3	Circulate; reveal and name what a wrong board most likely did.	Work it alone; boards up on the cue.
Whiteboards	11	One question at a time. Boards down until the cue, all up together; scan the back row first.	Answer on a board; hold it up on three.
Independent Set	6	Silent. Circulate and mark what you see; do not teach.	Six questions, written, alone.
IXL	5	Assign the day's IXL skills.	Start the skills; finish tonight.
Total	**53**		

6. Gradual Release

I do. The Notes slides and the worked half of each Example: the teacher writes, the class copies, and the OFF line is said out loud rather than printed.

We do. Each Example's Your Turn: the same steps on the student's own numbers, revealed and named a minute later.

You do. The nine-question whiteboard round (11 minutes) and then the six-question independent set, silent and written.

7. Higher-Order Questions

Question	DOK
A car goes 402.5 mi on a tank. How many tanks does a 1,207-mi trip take, and why is the answer a whole number?	DOK 2
Two rates are 12 km in 50 min and 7 mi in 45 min. Why does 12 > 7 tell you nothing yet?	DOK 3
Paint is mixed 2 : 4 : 5. If you know the white, how do you find the whole batch — and why is that a scale factor?	DOK 3

8. Checks for Understanding

each board, and what to do about it

#	What the board asks	Answer	If the board is wrong
1	A car does 28 miles per gallon. Its tank holds 15 gal. How far does it go on a full tank?	***420 mi***	1.87 — 28 ÷ 15 [Notes I: miles per gallon × gallons gives miles]; 43 — added
2	A printer prints 35 pages per minute. A job is 1,400 pages. How long does the job take?	***40 min***	49,000 — multiplied [Notes I: name what the step finds — pages × pages/min is not minutes]; 400 — place value
3	Mia earns $14.50 per hour at a shop 2 mi from home, 6 h a day for 5 days. How much does she earn?	***$435***	87 — one day's pay [Notes I: the question asks for five days]; 2,175 — 435 × 5, the days counted twice
4	A car does 30 miles per gallon with a 12-gal tank. It starts a 900-mi trip with a full tank. How many tanks of gas does the trip take?	***D — 3 tanks***	(A) 2.5 rounded down: half a tank of road left undriven [Notes I: a part of a tank is still a tank to buy]; (B) 900 ÷ 12: the gallons used as miles per tank [Notes I: the rate first]; (C) 900 ÷ 36: the two numbers added before dividing [B1G-M 7.AR.4.5: steps not named]
5	A cereal has 12 g of protein per 100 g. How many grams of protein in 1 kg?	***120 g***	12 — the rate copied with the wrong unit [Notes I: per 100 g, and a kilogram is ten of those]; 1,200 — × 100
6	1 mi = 1.61 km. Cyclist A rides 12 km in 50 min; cyclist B rides 7 mi in 45 min. Who is faster?	***C — cyclist B***	(A) 12 km and 12 km/h confused: a distance read as a speed [B1G-M 7.AR.4.5: a rate is per unit of time]; (B) 12 > 7, the distances compared in two units [Notes II: same unit first]; (D) 11.27 is B's distance in km, not a speed — the time was never used [B1G-M 7.AR.4.5: a rate is per unit of time]
7	Paint is mixed white : blue : grey = 2 : 4 : 5. A batch uses 8 gal of white. How many gallons is the whole batch?	***44 gal***	20 — 8 + 4 + 5 ×1, the ratio read as gallons [Notes III: k from the part you know]; 32 — the grey (5 × 4) and nothing more
8	A plan is drawn at 1 in : 4 ft; a room is 2 in by 2.5 in on it. 1 ft = 0.305 m. Find the room's area in square metres.	***About 7.44 m^2***	24.4 — 80 ft^2 × 0.305, the ratio once on an area [5.08–09 Notes II]; 20 — 2 × 2.5 × 4, the drawing's area scaled once [5.04 board 6]
9	1 mi = 1.61 km. A 300-km drive. Car A averages 50 mi/h; car B averages 75 km/h. Which car gets there first, and by how long? Explain.	***Car A: 80.5 km/h against 75, so about 3.73 h against 4 h — about 16 minutes sooner.***	'Car B, 75 > 50' — numbers compared in two units [Notes II: same unit first]; 'Car A, 80.5 > 75' with no times — the decision without the 'by how long'

MTR.4.1 — the one peer move. When a board splits — no option holding about two-thirds of the room — sixty seconds of “convince the person next to you”, then re-vote, then reveal. The minute comes out of the whiteboard remainder.

9. Differentiation

ESE / 504. Every board starts with the unit rate written as 'per 1' before anything is multiplied. For a comparison, a two-column table — one unit across the top — stays on the board, and nothing is compared until both columns are filled.

ELL. **Per** means for each one; **rate** is a per-sentence (miles per gallon, pages per minute). **Tank** on a board means a full tank of fuel.

Enrichment. Board 7 finishes early for some: ask for the batch if the grey were 25 gal instead (k = 5, 55 gal), and why the whole is always 11 × k.

10. Closure, Vocabulary, Materials, Practice

Closure. Board 9 is written work and it is the exit evidence: the two speeds in one unit (50 mi/h is 80.5 km/h), the comparison, and the decision in a sentence with its numbers. A board that picked a car without converting has compared numbers, not speeds.

Vocabulary. **unit rate**; **proportional**; **per**

Materials. Whiteboards and markers. The FAST reference sheet's conversion lines on the board (printed on every item as well). The Unit Review key to post. Calculators are allowed on everything and printed on nothing. 43 minutes: the timings below are the short ones.

Practice. The six-question independent set is worked silently in class; whatever is not finished goes home. IXL, all skills required to a SmartScore of 67, due at the start of the next class: Compare rates: word problems — VVS; Multi-step problems with customary or metric unit conversions — UCT.